

Power BI Star Schema Implementation Blueprint

Complete Technical Guide for Building the Exact Architectural Model

Star Schema Design

Power Query (M)

DAX Calendar

Relationship Cardinality

1. Executive Summary & Model Overview

This document details the step-by-step implementation process for structuring your Power BI dataset into an enterprise-grade Star Schema based on your architectural diagram. The data model enforces clear separation between dimension tables and fact tables, prevents direct fact-to-fact relationships, and ensures proper cardinality and cross-filtering rules.

Table Name	Original Source Name	Role in Model	Primary Key (PK)
DIM_product	raw_sku_catalog	Central Product Hub Dimension	parent_sku
BOM_data	raw_bom	Component Bill of Materials Dimension	bom_pk
FACT_Listings	cleaned_listings	Market Listings Fact Table	listing_id
FACT_Warranty_Listings	raw_warranty_claims	Warranty Claims Fact Table	claim_id
DIM_Date	DAX Calculated Table	Central Date Dimension	Date

2. Table Schema Definitions & Transformations

A. Primary Product Dimension (DIM_product)

Directly rename **raw_sku_catalog** to **DIM_product** in Power Query Editor (do not duplicate the table to avoid memory redundancy).

- **Columns:** parent_sku (PK), product_name, category, launch_date, original_price, discontinued, manufacturing_site.
- **Required Action:** Select parent_sku column → Right-click → **Remove Duplicates** to ensure exact 1-to-many relationship compatibility.

B. Bill of Materials Dimension (BOM_data)

Rename **raw_bom** to **BOM_data** and add the required composite primary key.

- **Columns:** bom_pk (PK), parent_sku (FK), parent_desc, component_sku, component_desc, unit_cost, supplier, warranty_months, mtbf_hours, component_type, qty_per_unit.
- **Creating Key Step (Power Query):** Go to **Add Column** → **Custom Column**:

```
Column Name: bom_pk
Formula: [parent_sku] & "-" & [component_sku]
```

C. Listings Fact Table (FACT_Listings)

Rename **cleaned_listings** to **FACT_Listings** and include required attributes.

- **Columns:** listing_id, parent_sku (or model_clean), confidence_level, brand_clean, condition, damage_flag, listing_url, location, price, scrape_date.
- **Required Actions:**
 1. Double-click the model/SKU matching column header and rename it to **parent_sku**.
 2. Ensure **confidence_level** is checked via **Choose Columns** or added as a custom field.

D. Warranty Claims Fact Table (FACT_Warranty_Listings)

Rename **raw_warranty_claims** to **FACT_Warranty_Listings**.

- **Columns:** claim_id, parent_sku (FK), component_sku (FK), component_type, failure_date, failure_reason, within_warranty.

3. DAX Date Dimension Table Generation

To avoid naming collision errors with ADDCOLUMNS and CALENDAR, create the **DIM_Date** table using the following corrected DAX expression:

```
DIM_Date =
ADDCOLUMNS (
    CALENDAR ( DATE(2020, 1, 1), DATE(2030, 12, 31) ),
    "Month", FORMAT([Date], "mmm"),
    "Month No", MONTH([Date]),
    "Quarter", "Q" & FORMAT([Date], "Q"),
    "Year", YEAR([Date])
)
```

DAX Resolution Note

The CALENDAR function automatically creates the base column named [Date]. Do not declare "Date", [Date] inside ADDCOLUMNS as it causes a duplicate column error.

4. Relationship Architecture & Cardinality Rules

Switch to **Model View** in Power BI Desktop to construct the star schema relationships:

From (1 - Dimension)	To (* - Fact / Child)	Join Key	Cardinality	Filter Direction
DIM_product	BOM_data	parent_sku \$ ightarrow\$ parent_sku	1 : *	Single
DIM_product	FACT_Listings	parent_sku \$ ightarrow\$ parent_sku	1 : *	Single
DIM_product	FACT_Warranty_Listings	parent_sku \$ ightarrow\$ parent_sku	1 : *	Single
BOM_data	FACT_Warranty_Listings	component_sku \$ ightarrow\$ component_sku	1 : *	Single
DIM_Date	FACT_Warranty_Listings	Date \$ ightarrow\$ failure_date	1 : *	Single
DIM_Date	FACT_Listings	Date \$ ightarrow\$ scrape_date	1 : *	Single

Architectural Constraints

- **No direct fact-to-fact joins:** Do not connect FACT_Listings directly to FACT_Warranty_Listings.
- **Filter flow:** All relationships must be configured as **Single** direction with filters originating from dimension tables (1 side) down to fact tables (* side).

5. Visual Navigation Guide for Power BI Desktop

1. **Opening Power Query Editor:** From the main ribbon, click **Home** \$ ightarrow\$ **Transform data** (under the *Queries* group).
2. **Applying Schema Updates:** Make column additions and table renames, then click **Close & Apply** on the far-left of the Home tab ribbon.
3. **Managing Model View:** Click the **Model View** icon on the far-left vertical sidebar, drag join columns between tables, and double-click relationship lines to verify Cardinality and Cross-filter direction settings.